



Finance for Non-Finance Managers

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Outline

- The different financial ratios
- Profitability Analysis
- Cash Flow Analysis
- Projected financial and cash-flow statements
- The process of financial statements analysis
- Use financial analysis to determine the health of an organisation
- Evaluate organisation's historical financial performance
- Investment suitability based on financial analysis
- Analysis for the financial services industry
- Benchmarking of financial performance against industry
- Use Capital Budgeting to evaluate potential investment returns

Financial Statements

- Collection of summary level reports
- Mainly used for covered
 - Financial Results
 - Financial Position
 - Cash Flows

Users of financial statements

Company Management

Competitors

Customers

Employees

Governments

Investment Analysts

Investors

Lenders

Rating Agencies

Suppliers

Key Financial Statements



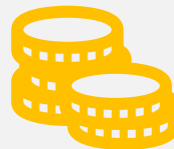
Statement of
Financial Position

Balance Sheet



Statement of
Financial Results

Profit and
Loss
Statement



Cash Flows

Cash Flows
statement

Balance Sheet

- Statement of net-worth or financial position
- Components
 - Assets
 - Liabilities
 - Equity
- Fundamental Equation
 - $\text{Assets} = \text{Liabilities} + \text{Equity}$

Balance Sheet Components

- Assets
 - Current Assets
 - Long-Term Assets
- Liabilities
 - Current Liabilities
 - Long-Term liabilities
- Total Shareholder's Equity
 - Share Capital
 - Retained Earnings

Assets

- Current Assets
 - Cash and cash equivalents
 - Accounts Receivable
 - Inventories
 - Marketable Securities
- Long-Term Assets
 - Plant Property and Equipment (PP&E)
 - Intangible Assets

Current Assets – Cash and Cash Equivalents

- Cash
 - Legal Tender
 - Bills
 - Coins
 - Checks
- Cash equivalents
 - Banker's Acceptance
 - Commercial Paper
 - Any short term investment securities with maturity less than 90 days

Cash vs Accrual basis accounting



Cash accounting records business transactions when cash flows into business



Accrual Accounting records business transactions when they are incurred regardless of when money actually changes hands



Financial Statements use accrual accounting

Current Assets – Accounts Receivable

- Funds owed by customers to the company
- The invoice amount is recorded as an Accounts Receivable once the company delivers goods or services to the customer.

Current
Assets -
Inventory

Merchandise

Raw materials

Work-in-progress

Finished products

Current Assets – Marketable Securities

- Common Stock
- Treasury Bills
- Bonds (only with maturity less than one year)

Long Term Assets – Plant, Property and Equipment (PP&E)

Machinery

Equipment

Vehicles

Buildings

Land

Office Equipment

Furniture

Fixtures

Valuation of PP&E

- Cost Model
- Revaluation Model

Depreciation of PP&E

- Spreads the cost of the asset over a period of time
- Match the expense of asset to income it produces
- The method of depreciation chosen can impact the financial results

Common Depreciation Methods

- Straight Line
- Double Declining
- Sum of Years
- Units of Production

Depreciation – Straight Line Method

- Expense amount is the same every year
- Depreciation expense = $(\text{Cost} - \text{Salvage Value}) / \text{Useful Life}$
- A machine is bought for \$50,000 and has useful life of 10 years and \$0 salvage value.
- Annual Depreciation expense = $(\$50,000 - \$0) / 10 = \$5000$

Straight Line Depreciation Illustration

Machine Purchase Value	\$50,000
Salvage Value	\$0
Annual Depreciation	\$5,000

Year	Book Value (Beginning of the Year)	Depreciation	Book Value (End of the Year)
1	\$50,000	\$5,000	\$45,000
2	\$45,000	\$5,000	\$40,000
3	\$40,000	\$5,000	\$35,000
4	\$35,000	\$5,000	\$30,000
5	\$30,000	\$5,000	\$25,000
6	\$25,000	\$5,000	\$20,000
7	\$20,000	\$5,000	\$15,000
8	\$15,000	\$5,000	\$10,000
9	\$10,000	\$5,000	\$5,000
10	\$5,000	\$5,000	\$0

Depreciation – Double Declining Method

Larger expense in earlier than later years

A machine is bought for \$50,000 and has useful life of 10 years and \$10,000 salvage value.

Depreciation rate = $(80\% / \text{useful life of asset}) \times 2 = 16\%$

Depreciation expense in first year = $\$50,000 \times 16\% = \$8,000$

Double Depreciation Balance Illustration

Machine Purchase Value	\$50,000
Salvage Value	\$10,000
Straight Line Depreciation Percentage	8%

Year	Book Value (Beginning of the Year)	Depreciation	Book Value (End of the Year)
1	\$50,000	\$8,000	\$42,000
2	\$42,000	\$6,720	\$35,280
3	\$35,280	\$5,645	\$29,635
4	\$29,635	\$4,742	\$24,894
5	\$24,894	\$3,983	\$20,911
6	\$20,911	\$3,346	\$17,565
7	\$17,565	\$2,810	\$14,755
8	\$14,755	\$2,361	\$12,394
9	\$12,394	\$1,983	\$10,411
10	\$10,411	\$1,666	\$8,745

The Depreciation expense in the 10th year will be adjusted to \$411 so that final value at the end of the period remains at \$10,000

Depreciation – Units of Production Method

- Assets are depreciated based on how long they have been used or the number of units produced
- Depreciation expense = (Number of units produced / Life number of units) x (Cost – salvage Value)
- A machine costs \$50,000 and an estimated production of 100 million and \$0 salvage value. During the first year, it produced 3 million units.
- Depreciation expense = (3 million / 100 million) x (\$50,000 - \$0) = \$1500

Depreciation - Units of Production Method

Machine Purchase Value	\$50,000
Salvage Value	\$0
Depreciable Value	\$50,000
Total Production (Million)	100

Year	Book Value (End of the Year)	Depreciable Value	Annual Production (Million)	Depreciation	Book Value (End of the Year)
1	\$50,000	\$50,000	3	\$1,500	\$48,500
2	\$48,500	\$50,000	5	\$2,500	\$46,000
3	\$46,000	\$50,000	2	\$1,000	\$45,000
4	\$45,000	\$50,000	6	\$3,000	\$42,000
5	\$42,000	\$50,000	3	\$1,500	\$40,500
6	\$40,500	\$50,000	4	\$2,000	\$38,500
7	\$38,500	\$50,000	6	\$3,000	\$35,500
8	\$35,500	\$50,000	3	\$1,500	\$34,000
9	\$34,000	\$50,000	10	\$5,000	\$29,000
10	\$29,000	\$50,000	4	\$2,000	\$27,000

Depreciation – Sum of Years Method

- Higher expense in early years and lower expense in latter years
- Depreciation expense = $(\text{Remaining Life} / \text{Sum of the years digits}) \times (\text{Cost} - \text{Salvage Value})$
- A machine is bought for \$25,000 and has life of 8 years with salvage value of \$0.
- Depreciation Expense
 - First Year - \$5556
 - Second Year - \$4861

Sum of Years Depreciation Illustration

Machine Purchase Value	\$50,000
Salvage Value	\$0
Sum of Years	55

Year	Book Value (Beginning of the Year)	Depreciation	Book Value (End of the Year)
1	\$50,000	\$9,091	\$40,909
2	\$40,909	\$8,182	\$32,727
3	\$32,727	\$7,273	\$25,455
4	\$25,455	\$6,364	\$19,091
5	\$19,091	\$5,455	\$13,636
6	\$13,636	\$4,545	\$9,091
7	\$9,091	\$3,636	\$5,455
8	\$5,455	\$2,727	\$2,727
9	\$2,727	\$1,818	\$909
10	\$909	\$909	\$0

Long Term Assets – Intangible Assets

Assets that cannot be touched

Goodwill

Brand recognition

Copyrights

Patents

Trademarks

Trade Names

Customer Lists

Liabilities

- Current Liabilities
 - Accounts Payable
 - Current Debt / Notes Payable
 - Current Portion of Long Term Debt
- Long-Term Liabilities
 - Bonds Payable
 - Long Term Debt

Current Liabilities – Accounts Payable

- Money owed to vendors or suppliers

Current Liabilities – Current Debt

- Formal borrowings of the company
- Payable within one year

Current Liabilities – Current Portion of Long Term Debt

- Unpaid principal from long term debt accrued within current operating cycle.

Long Term Liabilities – Bonds Payable

- Obligation to repay principal at a future date
- Repay interest on annual or semi-annual basis at regular intervals
- Types of bonds
 - Serial Bonds
 - Sinking Fund Bonds
 - Convertible Bonds
 - Registered Bonds
 - Secure Bonds
 - Debenture Bonds

Long Term Liabilities – Long Term Debt

- Debts owed to creditors payable beyond 12 months

Working Capital

- Current Assets – Current Liabilities

Shareholder's Equity

- Share Capital
- Retained Earnings

Shareholder's Equity – Share capital

- Money invested by owners in the business
- Can be common or preferred shares
- Common types of Share Capital
 - Authorized Share capital
 - Issued Share capital
 - Subscribed Capital
 - Called-up Capital
 - Paid-up Capital

Shareholders Equity – Retained Earnings

- Net Profits retained by the company
- Amount after all dividends and distributions

Income Statement

- Statement showing financial performance
- Also called “Profit and Loss Statement”
- Components
 - Revenues
 - Expenses
 - Profits
- Fundamental Equation
 - $\text{Profits} = \text{Revenues} - \text{Expenses}$

Income Statement Components

Revenue / Sales (-)

Cost of Goods Sold (=)

Gross Profit (-)

Marketing, Advertising Expenses (-)

General and Administrative Expenses (=)

Earnings Before Interest Taxes Depreciation Amortization EBITDA (-)

Depreciation and Amortization Expenses (=)

Operating Income or EBIT (-)

Interest Expenses (-)

Other Expenses (=)

EBT or Earnings Before Tax (-)

Income Tax (=)

Net Income

Layout of the Income Statement



COMPANY B INCOME STATEMENT

For Year Ended September 28, 2019 (In thousands)

NET SALES	\$ 4,358,100
COST OF SALES	2,738,714
GROSS PROFIT	1,619,386
SELLING AND OPERATING EXPENSES	560,430
GENERAL AND ADMINISTRATIVE EXPENSES	293,729
TOTAL OPERATING EXPENSES	854,159
OPERATING INCOME	765,227
OTHER INCOME	960
GAIN (LOSS) ON FINANCIAL INSTRUMENTS	5,513
(LOSS) GAIN ON FOREIGN CURRENCY	(12,649)
INTEREST EXPENSE	(18,177)
INCOME BEFORE TAXES	740,874
INCOME TAX EXPENSE	257,642
NET INCOME	\$ 483,232

Summary of Income Statement Items

- Revenue – Cost of Goods Sold = Gross Profit
- Gross Profit – Marketing expenses – Selling and General Admin Expenses = Earnings Before Interest Tax Depreciation and Amortization (EBITDA)
- EBITDA – Depreciation – Amortization = Operating Income or Earnings Before Interest and Tax (EBIT)
- EBIT – Interest and Other expenses = Earnings Before Tax (EBT)
- EBT – Tax = Net Income

Cash Flow Statement

- Shows the movement of cash in the business
- Tracks incoming and outgoing money
- Three cash activities of the company
 - Operating Activities
 - Investing Activities
 - Financial Activities

Cashflow Statement – Operating Activities

- Generates revenue for the company (inflow)
- Money spent on production of goods (outflow)
- Operating Cashflow = Operating Income + Depreciation – Taxes + Change in Working Capital

Cashflow Statement – Investing Activities

- Gains and Losses from investments in Assets

Cashflow Statement – Financing Activities

- Cash flow between company, shareholders and creditors

Example of a cashflow statement



Cash Flow Statement For the Year Ended December 31, 2016	
Cash Flow from Operations	
Net income	79,000
Adjustments for depreciation	2,000
Adjustments for increase in inventories	(22,000)
Adjustments for decrease in accounts receivable	12,000
Net Cash Flow from Operations	71,000
Cash Flow from Investing	
Cash receipts from sale of property and equipment	10,000
Cash paid for purchase of equipment	(12,000)
Net Cash Flow from Investing	(2,000)
Cash Flow from Financing	
Cash paid for loan repayment	(5,500)
Net Cash Flow from Financing	(5,500)
Net Increase in Cash	63,500

Summary of Cash Flow items

- Operating Cashflows
 - Income from sale of goods
 - Expenses in production of goods
- Investing Cashflows
 - Gains or Losses from sale or purchase of assets
- Financing Cashflows
 - Inflow or outflow from capital or loans

Statement of changes in Equity

- Also called the statement of Retained Earnings
- Reconciles Beginning and Ending equity during reporting period
- $\text{Beginning Equity} + \text{Net Income} - \text{Dividends} \pm \text{Other changes} = \text{Ending Equity}$

Statement of changes in Equity - components

- Opening Balance
- Effects of changes in accounting policies
- Effect of correction in prior period error
- Restated Balance
- Changes in Share Capital
- Dividends
- Income/Loss for the Period
- Changes in Revaluation Reserve
- Other Gains and Losses

Financial Ratios



LIQUIDITY
RATIOS



LEVERAGE
RATIOS



EFFICIENCY
RATIOS



PROFITABILITY
RATIOS

Liquidity Ratios

Current Ratio

Acid Test Ratio

Cash Ratio

Operating cash flow ratio

Liquidity Ratios – Current Ratio

Also called Working Capital Ratio

Current Ratio = Current Assets / Current Liabilities

Liquidity Ratios – Acid Test Ratio

Also called Quick Ratio

Acid Test Ratio = $(\text{Current Assets} - \text{Inventories}) / \text{Current Liabilities}$

Liquidity Ratios – Cash Ratio

Cash Ratio= Cash and Cash Equivalents / Current Liabilities

Liquidity Ratios – Operating Cash Flow Ratio

Operating Cash Flow Ratio = Operating Cash Flow / Current Liabilities

Leverage Financial Ratios

- Debt Ratio
- Debt to Equity Ratio
- Interest Coverage Ratio
- Debt Service Coverage Ratio

Leverage Ratio – Debt Ratio

Debt Ratio = Total liabilities / Total assets

Leverage Ratio – Debt to Equity Ratio

Debt to equity ratio = Total liabilities / Shareholder's equity

Leverage Ratio – Interest Coverage Ratio

Interest coverage ratio = Operating income / Interest expenses

Leverage Ratio – Debt Service Coverage Ratio

Debt service coverage ratio = Operating income / Total debt service

Efficiency Ratios

- Asset Turnover Ratio
- Inventory Turnover Ratio
- Receivables Turnover Ratio
- Payables Turnover Ratio

Efficiency Ratio – Asset Turnover Ratio

Asset turnover ratio = Net sales / Average total assets

Efficiency Ratio – Inventory Turnover Ratio

Inventory turnover ratio = Cost of goods sold / Average inventory

Efficiency Ratio – Accounts Receivable Turnover Ratio

Receivables turnover ratio = Net sales / Average accounts receivable

Average Collection Period

- Average Length of time for a company to collect the accounts receivable
- Average Collection Period = $365 / \text{Accounts Receivables Turnover}$

Importance of Average Collection Period

- Maintain Liquidity
- Plan for future costs and schedule potential expenditures

Average Collections Period - Illustration

A company has accounts receivable balance of \$40,000 in the balance sheet. For the same year, the total net sales is \$400,000.

$$\text{Average collection Period} = 365 \times 40,000 / 400,000 = 36.5$$

Efficiency Ratio – Accounts Payable Turnover Ratio

Payables turnover ratio = Net purchases / Average accounts payable

Net purchases = Cost of Goods Sold + Ending Inventory Balance – Starting Inventory Balance

Average Payments Period

Average time taken for a business to pay its vendors.

Average Payments Period = $365 / \text{Accounts Payable Turnover}$

Average Payments Period - Illustration

For a company

Beginning Accounts Payable : \$100,000

Ending Accounts Payable : \$ 80,000

Total Purchases on credit : \$1,800,000

Average Accounts Payable : $(\$100,000 + \$80,000) / 2 = \$90,000$

Accounts Payable Turnover : $\$1,800,000 / \$90,000 = 2$

Accounts Payable Turnover = $365 / 2 = 182.50$



Profitability Ratios

- Reflect ability of business to generate earnings
- Show efficiency of generating profits
- Higher ratios are more favourable

Profitability Ratios

Gross Margin

Operating Margin

Net Profit Margin

Return on Assets

Return on Equity

Profit Ratio – Gross Margin

Gross margin ratio = Gross profit / Net sales

Profit Ratio – Operating Margin

Operating margin ratio = Operating income / Net sales

Profit Ratio – Net Profit Margin

Net Profit Margin ratio = Net Income / Net sales

Profit Ratio – Return on Assets

Return on assets ratio = Net income / Total assets

Profit Ratio – Return on Equity

Return on equity ratio = Net income / Total Equity

How do we measure profitability

Balance Sheet (All figures in '000)	
Assets	
Current Assets	1000
Long Term Assets	1500
Total Assets	2500
Liabilities	
Current Liabilities	500
Long Term Liabilities	1400
Total Liabilities	1900
Equity	
Shareholder Equity	400
Retained Earnings	200
Total Equity	600

Income Statement (All figures in '000)	
Sales	200
Cost of Goods Sold	80
Gross Profit	120
Marketing Expenses	20
General and Admin Exp	25
Operating Profit (EBIT)	75
Interest Expenses	10
Income Before Tax	65
Taxes	13
Net Income	52

Gross Profit**Margin**

Gross Profit / Sales 60%

Operating Profit**(EBITDA) Margin**

Operating Profit /
Sales 38%

EBT Margin

Income Before Tax /
Sales 33%

Net Profit Margin

Net Income / Sales 26%

Return on Assets

Net Income / Total
Assets 2%

Return on Equity

Net Income /
Shareholder Equity 13%

Summary of all ratios

- Liquidity Ratios
 - Current Ratio = Current Assets / Current Liabilities
 - Acid Test Ratio = (Current Assets – Inventories) / Current Liabilities
 - Cash Ratio = Cash and Cash Equivalents / Current Liabilities
 - Operating Cash Flow Ratio = Operating Cash Flow / Current Liabilities
- Leverage Ratios
 - Debt Ratio = Total liabilities / Total assets
 - Debt to equity ratio = Total liabilities / Shareholder's equity
 - Debt service coverage ratio = Operating income / Total debt service

Summary of all ratios

- Efficiency Ratios
 - Asset turnover ratio = $\text{Net sales} / \text{Average total assets}$
 - Inventory turnover ratio = $\text{Cost of goods sold} / \text{Average inventory}$
 - Receivables turnover ratio = $\text{Net sales} / \text{Average accounts receivable}$
 - Payables turnover ratio = $\text{Net purchases} / \text{Average accounts payable}$
- Profitability Ratios
 - Gross margin ratio = $\text{Gross profit} / \text{Net sales}$
 - Operating margin ratio = $\text{Operating income} / \text{Net sales}$
 - Net Profit Margin ratio = $\text{Net Income} / \text{Net sales}$
 - Return on equity ratio = $\text{Net income} / \text{Total Equity}$
 - Return on assets ratio = $\text{Net income} / \text{Total assets}$

Profitability Ratios for a sample company

- Gross Margin : 37.50%
- Operating Margin : 26.67%
- Net Profit Margin : 16.67%
- Return on Equity : 9.09%
- Return on Assets : 4%

Sample Balance Sheet of Company A	
	200
	300
	500
	30
	250
	280
	100
	120
	220

Summary of Income Statement	
(All figures in '000)	
Sales	
Cost of Goods Sold	
Gross Profit	
Marketing Expenses	
General Administrative Expenses	
Operating Profit	
Interest Expense	
Income Before Taxes	
Taxes	
Net Income	

Cash Flow Analysis

1

Step 1 : Identify all sources of income

2

Step 2 : Identify all business expenses

3

Step 3 : Create the Cash Flow Statement

- Operating Activities
- Investing Activities
- Financing Activities

4

Step 4 : Analyse the cash flow statement

- Look for any deficit patterns
- Better time the expenses

How can business improve cash flow management

1

Plan for upcoming expenses

2

Increase the Income

3

Re-examine payment schedules

- Look at average collections period
- Look at average payables period

Example of a monthly cash flow statement

ABC Cleaning Service			
Statement of Cash Flows			
For the month ended January 31, 2020			
CASH AND CASH EQUIVALENTS, BEGINNING OF PERIOD			\$ 50,000.00
OPERATING ACTIVITIES			
	Net Income		\$ 6,475.00
	ADJUST FOR OPERATING ACTIVITIES:		
	Depreciation on fixed assets		\$ 1,000.00
	Decrease in current assets:		
	- Accounts receivable		\$ (150.00)
	- Inventory		\$ (19,900.00)
	- Prepaid expenses		\$ (500.00)
	Increase in current liabilities:		
	- Accounts payable		\$ 24,500.00
	- Deferred revenue		\$ 725.00
	NET CASH FROM OPERATING ACTIVITIES		\$ 12,150.00

INVESTING ACTIVITIES			
	Proceeds from marketable securities		\$ 10,158.00
	Purchase of property or equipment		\$ (5,500.00)
	Payments related to business acquisitions		\$ (248.00)
	Payments related to property acquisitions		\$ (4,082.00)
	Payments related to intangible asset acquisitions		\$ (455.00)
	NET CASH FROM INVESTING ACTIVITIES		\$ (127.00)
FINANCING ACTIVITIES			
	Proceeds from line of credit		\$ 1,000.00
	Proceeds from long-term debt		\$ 49,750.00
	Payments on long-term debt		\$ (20,000.00)
	Payments on line of credit		\$ (10,000.00)
	Repurchase of common stock		\$ (11,430.00)
	NET CASH FROM FINANCING ACTIVITIES		\$ 9,320.00
	Increase/Decrease in cash and cash equivalents		\$ 21,343.00
	CASH AND CASH EQUIVALENTS, END OF PERIOD		\$ 71,343.00

Proforma Financial Statements

Projected Financial Statements

- Help in make plans and compare actual and predicted company performance
- Projection done for Income Statement and Balance Sheet
- Projected Cash Flow is derived from the other statements

Forecasting of Income Statement

1

Analyse
Historical
Data

2

Forecast
Revenue

3

Predict Cost
of Goods
Sold

4

Determine
Operating
Expenses

5

Calculate
Projected Net
Income

Forecasting a Balance Sheet

1

Enter Short term
and Long term
assets

2

Include current and
Long Term Liabilities

3

Calculate the Final
Figures



Forecasting the Cash Flow Statement

Estimate the amount of cash inflows and outflows

Steps

- Estimated anticipated monthly sales
- Predict receipt of payments
- Estimate Costs



Direct

Cash flow =
receivables -
expenditures



Indirect

Start with net
income and
account for
items affecting
profit but not
cash flow

Determine financial health of a company

Analyse Balance Sheet

Analyse Income
Statement

Analyse Cash Flow
Statement

Financial Ratio
Analysis

Determine Financial Health of a company

Analyse the Balance Sheet

- How much debt the company has relative to equity
- How liquid the business is in the short term (less than one year)
- What percentage of assets are tangible and what percentage comes from financial transactions
- How long it takes to receive outstanding payments from customers and repay suppliers
- How long it takes to sell inventory the business keeps on hand

Determine Financial Health of a company

Analyse the Income Statement

- How much revenue is growing over certain accounting periods
- The gross profit margin for goods sold
- What percentage of revenue results in net profit after all expenses
- If the business can cover its interest repayments on debt
- How much the business repays to shareholders versus how much it reinvests

Determine Financial Health of a company

Analyse the Cash Flow Statement

- The liquidity situation of the company
- The company's sources of cash
- The free cash flow the company generates to further invest in assets or operations
- Whether overall cash has increased or decreased

Determine Financial Health of a company

Financial Ratio Analysis

- **Gross profit margin:** The percentage of profit the company generates after direct cost of sales expenses have been deducted from the revenue
- **Net profit margin:** The percentage of profit the company generates after all expenses have been deducted from revenue, including interest and tax from revenue
- **Coverage ratio:** The company's ability to meet its financial obligations, specifically to cover its debt and related interest payments
- **Current ratio:** The company's ability to meet short-term obligations of less than one year
- **Quick ratio:** The company's ability to meet short-term obligations of less than one year using only highly liquid assets

Determine Financial Health of a company

Financial Ratio Analysis

- **Debt-to-equity ratio:** The percentage of debt versus equity that the company uses to finance itself
- **Inventory turnover:** How many times per period the entire inventory was sold
- **Total asset turnover:** How efficiently the company generates revenue from total assets
- **Return on equity (ROE):** The company's ability to use equity investments to earn profit
- **Return on assets (ROA):** The company's ability to manage and use its assets to earn profit

Financial Statements Analysis

- Evaluate the company's financial situation
- Follow up in administrative decisions
- Assist in economic decisions





Financial Statements Analysis - Methods

- Ratio Analysis
- Horizontal Analysis (Trend Analysis)
- Vertical Analysis

Ratio Analysis

- Using the ratios of a business to evaluate its financial position

Balance Sheet of company A and B

<u>Assets</u>	<u>Company A</u>	<u>Company B</u>
Current Assets		
Cash	\$30,000	\$30,000
Inventory	\$30,000	\$60,000
Accounts Receivable	\$70,000	\$140,000
Fixed Assets		
Property	\$140,000	\$210,000
Others	\$40,000	\$50,000
Total	\$310,000	\$490,000

Liabilities		
Current Liabilities		
Accounts Payable	\$70,000	\$70,000
Short Term Loans	\$35,000	\$70,000
Long Term Liabilities	\$70,000	\$140,000
Capital	\$135,000	\$210,000
Total	\$310,000	\$490,000

Income Statement of company A and B

<u>Item</u>	<u>Company A</u>	<u>Company B</u>
Sales	\$140,000	\$210,000
Cost of Goods Sold	\$70,000	\$105,000
Gross Profit	\$70,000	\$105,000
Selling and Distribution Expenses	\$15,000	\$25,000
Earning Before Interest and Tax (EBITDA)	\$55,000	\$80,000
Depreciation	\$5,000	\$5,000
Earning Before Interest and Tax (EBIT)	\$50,000	\$75,000
Interest	\$5,000	\$10,000
Earning Before Tax	\$45,000	\$65,000
Taxes	\$9,000	\$13,000
Net Income	\$36,000	\$52,000

Ratio Analysis - example

Current Ratio = Current Assets / Current Liabilities

Item	<u>Company A</u>	<u>Company B</u>
Current Assets	\$130,000	\$230,000
Current Liabilities	\$105,000	\$140,000
Current Ratio	1.24	1.64

Current Ratio

Quick Ratio= (Current Assets-Inventory) / Current Liabilities

Item	<u>Company A</u>	<u>Company B</u>
Quick Assets	\$100,000	\$170,000
Current Liabilities	\$105,000	\$140,000
Current Ratio	0.95	1.21

Quick Ratio

Operating Profit = Earnings Before Interest & Tax

Item	<u>Company A</u>	<u>Company B</u>
Operating Profit	\$50,000	\$75,000
Net Sales	\$140,000	\$210,000
Operating Profit Ratio	36%	36%

Operating
Profitability

Net Profit Ratio = Earnings Before Interest & Tax,

Item	<u>Company A</u>	<u>Company B</u>
Net Profit	\$36,000	\$52,000
Net Sales	\$140,000	\$210,000
Net Profit Ratio	26%	25%

Net Profit Ratio

Return on Equity = Net Profit / Shareholders Equity

Item	<u>Company A</u>	<u>Company B</u>
Net Profit	\$36,000	\$52,000
Equity	\$135,000	\$210,000
Return on Equity	27%	25%

Return on Equity

Return on Capital Employed = Earnings before Interest / Capital Employed

Item	<u>Company A</u>	<u>Company B</u>
EBIT	\$50,000	\$75,000
Capital Employed	\$205,000	\$350,000
Return on Capital Employed	24%	21%

Return on Capital Employed

Inventory Turnover

Inventory Turnover Ratio = Cost of Goods Sold / Average Inventory

Item	<u>Company A</u>	<u>Company B</u>
Cost of Goods Sold	\$70,000	\$105,000
Average Inventory	\$30,000	\$60,000
Inventory Turnover Ratio	2.33	1.75

Receivable Turnover

Receivables Turnover Ratio = Credit Sales / Average Receivables

Item	<u>Company A</u>	<u>Company B</u>
Credit Sales	\$140,000	\$210,000
Average Receivables	\$70,000	\$140,000
Receivables Turnover Ratio	2.00	1.50

Payable Turnover

Payables Turnover Ratio = Credit Sales / Average Receivables

Assume 50% COGS is purchases

Item	<u>Company A</u>	<u>Company B</u>
Purchases	\$35,000	\$52,500
Average Payables	\$70,000	\$70,000
Payables Turnover Ratio	0.50	0.75

Debt to Equity Ratio

Debt to Equity Ratio = Total Long Term Debt / Total Equity

Item	<u>Company A</u>	<u>Company B</u>
Long Term Debt	\$70,000	\$140,000
Equity	\$135,000	\$210,000
Payables Turnover Ratio	0.52	0.67

Financial Leverage

Financial Leverage = Total Assets / Total Equity

Item	<u>Company A</u>	<u>Company B</u>
Total Assets	\$310,000	\$490,000
Equity	\$135,000	\$210,000
Payables Turnover Ratio	2.30	2.33

Horizontal Analysis

- Analyse change over two or more accounting periods
- Measure percentage change of same line item

Horizontal Analysis - Example

Safeway Stores Comparative Balance Sheets				
			Increase or Decrease	
Assets	2019	2018	Amount	Percentage
Current Assets:	\$	\$	\$	%
Cash and short-term investments	74,442	75,574	(1,132)	(1.50)
Receivables	105,166	91,642	13,524	14.76
Merchandise inventories	1,563,244	1,433,254	129,990	9.07
Prepaid expenses and other current assets	118,537	128,676	(10,139)	(7.88)
Total Current Assets	1,861,389	1,729,146	132,243	7.65
Property				
Land	236,876	210,427	26,449	12.57
Buildings	345,001	305,006	39,995	13.11
Leasehold improvements	557,504	459,711	97,793	21.27
Fixtures and equipment	2,023,914	1,789,136	234,778	13.12
Transport equipment	186,485	173,576	12,909	7.44
Property under capital lease	1,144,409	1,155,493	(11,084)	(0.96)
	4,494,189	4,093,349	400,840	9.79
Less: Accumulated depreciation and ammortization	1,894,333	1,731,138	163,195	9.43
Total property, net	2,599,856	2,362,211	237,645	10.06
Excess of cost over net assets acquired	16,588	16,557	31	0.19
Other Assets	59,396	66,449	(7,053)	(10.61)
Total Assets	4,537,229	4,174,363	362,866	8.69

Vertical Analysis

- Each Item in the financial statement is shown in percentage of the base figure
- All items in income statement are stated as percentage of gross sales
- All items in balance sheet are stated as percentage of total assets



Vertical Analysis - Example

ABC Company Vertical Analysis

	Year 1	Year 2	Year 3
Sales	100%	100%	100%
Cost of Goods Sold	30%	29%	40%
Gross Profit	70%	71%	60%
Salaries	24%	23%	30%
Rent	6%	6%	5%
Marketing	5%	5%	10%
Utilities	1%	1%	1%
Other expenses	2%	2%	2%
Total Expenses	38%	37%	48%
Net Income	32%	34%	12%

ITEM	YEAR			
<u>Assets</u>	<u>2021</u>	<u>2020</u>	<u>2019</u>	<u>2018</u>
Current Assets				
Cash	\$30,000	\$30,000	\$30,000	\$30,000
Inventory	\$40,000	\$34,000	\$32,000	\$30,000
Accounts Receivable	\$88,180	\$81,648	\$75,600	\$70,000
Fixed Assets				
Property	\$194,000	\$196,000	\$198,000	\$200,000
Others	\$39,998	\$39,999	\$40,000	\$40,000
Total	\$392,178	\$381,647	\$375,600	\$370,000

Liabilities				
Current Liabilities				
Accounts Payable	\$80,000	\$60,000	\$51,000	\$50,000
Short Term Loans	\$44,090	\$40,824	\$37,800	\$35,000
Long Term Liabilities	\$50,000	\$50,000	\$50,000	\$50,000
Capital and Retained Earnings	\$218,088	\$230,823	\$236,800	\$235,000
Total	\$392,178	\$381,647	\$375,600	\$370,000

ITEM	YEAR			
	<u>2021</u>	<u>2020</u>	<u>2019</u>	<u>2018</u>
Sales	\$186,340	\$169,400	\$154,000	\$140,000
Cost of Goods Sold	\$81,034	\$77,175	\$73,500	\$70,000
Gross Profit	\$105,306	\$92,225	\$80,500	\$70,000
Selling and Distribution Expenses	\$15,000	\$15,000	\$15,000	\$15,000
Earning Before Interest and Tax (EBITDA)	\$90,306	\$77,225	\$65,500	\$55,000
Depreciation	\$2,000	\$2,000	\$2,000	\$2,000
Earning Before Interest and Tax (EBIT)	\$88,306	\$75,225	\$63,500	\$53,000
Interest	\$4,997	\$4,998	\$4,999	\$5,000
Earning Before Tax	\$83,309	\$70,227	\$58,501	\$48,000
Taxes	\$16,662	\$14,045	\$11,700	\$9,600
Net Income	\$66,647	\$56,182	\$46,801	\$38,400

Horizontal Analysis – Case Study

ITEM	YEAR			
<u>Ratios</u>	<u>2021</u>	<u>2020</u>	<u>2019</u>	<u>2018</u>
<i>Liquidity Ratios</i>				
Current Ratio	1.27	1.44	1.55	1.53
Acid Ratio	0.95	1.11	1.19	1.18
<i>Efficiency Ratios</i>				
Asset Turnover	0.48	0.45	0.41	0.38
Inventory Turnover	5.04	5.13	4.97	4.67
Payables Turnover	0.58	0.70	0.73	0.70
Receivables Turnover	2.19	2.15	2.12	2.00
*Assume 50% of COGS are purchases				
<i>Leverage Ratios</i>				
Debt to Equity Ratio	0.80	0.65	0.59	0.57
Debt to Assets Ratio	0.44	0.40	0.37	0.36
<i>Profitability Ratios</i>				
Operating Margin	47%	44%	41%	38%
Net Profit Margin	36%	33%	30%	27%

Step 1 – Generation of Ratios

ITEM	YEAR						
<u>Assets</u>	<u>2021</u>	<u>YoY Growth</u>	<u>2020</u>	<u>YoY Growth</u>	<u>2019</u>	<u>YoY Growth</u>	<u>2018</u>
Current Assets							
Cash	\$30,000	0.00%	\$30,000	0.00%	\$30,000	0.00%	\$30,000
Inventory	\$40,000	17.65%	\$34,000	6.25%	\$32,000	6.67%	\$30,000
Accounts Receivable	\$88,180	8.00%	\$81,648	8.00%	\$75,600	8.00%	\$70,000
Fixed Assets							
Property	\$194,000	-1.02%	\$196,000	-1.01%	\$198,000	-1.00%	\$200,000
Others	\$39,998	0.00%	\$39,999	0.00%	\$40,000	0.00%	\$40,000
Total	\$392,178	2.76%	\$381,647	1.61%	\$375,600	1.51%	\$370,000
Liabilities							
Current Liabilities							
Accounts Payable	\$80,000	33.33%	\$60,000	17.65%	\$51,000	2.00%	\$50,000
Short Term Loans	\$44,090	8.00%	\$40,824	8.00%	\$37,800	8.00%	\$35,000
Long Term Liabilities	\$50,000	0.00%	\$50,000	0.00%	\$50,000	0.00%	\$50,000
Capital and Retained Earnings							
	\$218,088	-5.52%	\$230,823	-2.52%	\$236,800	0.77%	\$235,000
Total	\$392,178	2.76%	\$381,647	1.61%	\$375,600	1.51%	\$370,000

Step 2 – Trend Analysis of Balance Sheet

ITEM	YEAR						
	<u>2021</u>	<u>YoY Growth</u>	<u>2020</u>	<u>YoY Growth</u>	<u>2019</u>	<u>YoY Growth</u>	<u>2018</u>
Sales	\$186,340	10.00%	\$169,400	10.00%	\$154,000	10.00%	\$140,000
Cost of Goods Sold	\$81,034	5.00%	\$77,175	5.00%	\$73,500	5.00%	\$70,000
Gross Profit	\$105,306	14.18%	\$92,225	14.57%	\$80,500	15.00%	\$70,000
Selling and Distribution Expenses	\$15,000	0.00%	\$15,000	0.00%	\$15,000	0.00%	\$15,000
Earning Before Interest and Tax (EBITDA)	\$90,306	16.94%	\$77,225	17.90%	\$65,500	19.09%	\$55,000
Depreciation	\$2,000	0.00%	\$2,000	0.00%	\$2,000	0.00%	\$2,000
Earning Before Interest and Tax (EBIT)	\$88,306	17.39%	\$75,225	18.46%	\$63,500	19.81%	\$53,000
Interest	\$4,997	-0.02%	\$4,998	-0.02%	\$4,999	-0.02%	\$5,000
Earning Before Tax	\$83,309	18.63%	\$70,227	20.04%	\$58,501	21.88%	\$48,000
Taxes	\$16,662	18.63%	\$14,045	20.04%	\$11,700	21.88%	\$9,600
Net Income	\$66,647	18.63%	\$56,182	20.04%	\$46,801	21.88%	\$38,400

Step 3 – Trend Analysis of Income Statement

ITEM	YEAR						
<u>Ratios</u>	<u>2021</u>	<u>YoY Growth</u>	<u>2020</u>	<u>YoY Growth</u>	<u>2019</u>	<u>YoY Growth</u>	<u>2018</u>
<i>Liquidity Ratios</i>							
Current Ratio	1.27	-11.76%	1.44	-6.77%	1.55	1.32%	1.53
Acid Ratio	0.95	-14.00%	1.11	-6.88%	1.19	1.08%	1.18
<i>Efficiency Ratios</i>							
Asset Turnover	0.48	7.64%	0.45	8.31%	0.41	9.17%	0.38
Inventory Turnover	5.04	-1.89%	5.13	3.33%	4.97	6.45%	4.67
Payables Turnover	0.58	-16.75%	0.70	-4.46%	0.73	3.96%	0.70
Receivables Turnover	2.19	1.85%	2.15	1.85%	2.12	5.77%	2.00
*Assume 50% of COGS are purchases							
<i>Leverage Ratios</i>							
Debt to Equity Ratio	0.80	22.17%	0.65	11.48%	0.59	2.03%	0.57
Debt to Assets Ratio	0.44	12.33%	0.40	6.94%	0.37	1.28%	0.36
<i>Profitability Ratios</i>							
Operating Margin	0.47	6.72%	0.44	7.70%	0.41	8.92%	0.38
Net Profit Margin	0.36	7.84%	0.33	9.13%	0.30	10.80%	0.27

Trend Analysis of Ratios

Financial Analysis to determine investment suitability

Revenues

- Quantity, quality and timing of revenues
- Revenue Growth (ignore one-time revenues)
- Revenue Concentration
- Revenue per employee

Financial Analysis to determine investment suitability

Profits

- Gross Profit Margin
- Operating Profit Margin
- Net Profit Margin

Financial Analysis to determine investment suitability

Operational Efficiency

- Accounts receivables Turnover
- Inventory Turnover

Market Data

- Price to Earnings (PE) ratio
- Price to Book (PB) ratio
- Price Earnings Growth (PEG) Ratio

Financial Analysis to determine investment suitability

Capital Efficiency and Solvency

- Return on Equity
- Debt to Equity

Liquidity

DuPoint Index

Net Profit Margin x Asset Turnover x Equity Multiplier

$(\text{Net Income} / \text{Revenue}) \times (\text{Sales} / \text{Total Assets}) \times (\text{Total Assets} / \text{Shareholders Equity})$

Financial Ratios in Banking

Profitability Calculation Ratios

- Net Interest Margin : $(\text{Interest Income} - \text{Interest Expense}) / \text{Total Assets}$

Efficiency Calculation Ratios

- Efficiency Ratio : $\text{Non-Interest Expense} / \text{Revenue}$
- Operating Leverage : $\text{Growth Rate of Revenue} - \text{Growth Rate of Non-Interest Expense}$

Financial Ratios in Banking

Financial Strength Calculation Ratios

- Liquidity Coverage Ratio: $\text{High Quality Liquid Asset Amount} / \text{Total Net Cash Flow Amount}$
- Leverage Ratio : $\text{Tier 1 Capital} / \text{Total Assets (Exposure)}$
- CET1 Ratio : $\text{Common Equity Tier 1 Capital} / \text{Risk-Weighted Assets}$

Financial Ratios in Insurance

- Persistency Ratio

Number of policyholders paying premium / Net Active policyholders

- Solvency Ratio

Available Solvency margin / Required Solvency Margin

- Combined Ratio

Total outflow on operating expenses, commissions, claims and losses / Net Earned Premium

Financial Ratios in Insurance

- Incurred Claims Ratio

Total Claims paid / Total premiums collected

- Commission expense ratio

Commission Payment / Premiums Collected

- Claim Settlement Ratio

Number of claims paid out / Number of claims received

Benchmarking of Financial performance

Compare ratios of the company with leading competitors

Compare ratios with industry average

Benchmarking of Financial performance

Illustration of trend of US sectoral averages

Financial ratio	2021	2020	2019	2018	2017	2016
Solvency Ratios						
Debt ratio	0.57	0.61	0.61	0.62	0.63	0.65
Debt-to-equity ratio	0.93	1.07	1.12	0.87	0.80	0.75
Interest coverage ratio	3.04	1.22	2.20	1.98	1.84	1.69
Liquidity Ratios						
Current Ratio	2.03	1.94	1.69	1.58	1.56	1.53
Quick Ratio	1.25	1.25	1.08	1.05	1.09	1.09
Cash Ratio	0.82	0.82	0.51	0.40	0.42	0.41
Profitability Ratios						
Profit margin	5.2%	1.5%	4%	3.3%	2.6%	1.9%
ROE (Return on equity), after tax	2.6%	-2.1%	2.7%	0.7%	-1.1%	-2.5%
ROA (Return on assets)	1.1%	-0.1%	1%	0.8%	0.5%	0.2%
Gross margin	41.6%	41.5%	42.2%	42.2%	41.7%	41%
Operating margin (Return on sales)	9.7%	4.8%	8%	7.4%	6.4%	5.7%
Activity Ratios						
Asset turnover (days)	815	869	747	654	635	636
Receivables turnover (days)	48	54	50	48	48	48
Inventory turnover (days)	76	79	71	71	70	71
Price Ratios						
Dividend Payout Ratio	0.27	0.27	0.32	0.27	0.31	0.29

Capital Budgeting

- Planning for long term investments
- Focus on cash flows rather than profits
- Look at cash inflow and outflow and not accounting revenue and expenses
- Includes adjustments for the time value of money

Time Value of Money

- Money in hand is more valuable than same amount of money in the future
- Opportunity cost is the interest that can be earned on the money or due to inflation
- Components
 - Present Value (PV)
 - Future Value (FV)
 - Number of Time Periods (N)
 - Interest Rate (I)
 - Payment Amount (PMT)

Present Value and Future Value

$$PV = FV / [1 + i] ^ n$$

$$FV = PV \times [1 + i] ^ n$$

PV = Present Value

FV = Future Value

i= Interest Rate

n= Number of years

Present Value and Future Value - Example

- Interest rate = 5%
- Future Value of \$10,000 today after 2 years
- $\$10,000 \times (1 + 5\%)^2 = \$11,025$
- Present Value of \$10,000 to be received after 2 years
- $\$10,000 / (1 + 5\%)^2 = \9070.30

Capital Budgeting Process

- Identify long term goals
- Identify potential investment proposals
- Estimate and analyse the relevant cash flows from the proposal
- Analyse financial feasibility
- Choose projects to implement
- Implement the projects
- Monitor the projects

Capital Budgeting Methods

Payback Period

Discounted Payback Period

Net Present Value

Profitability Index

Internal Rate of Return

Payback Period



Time to recover the original investment



Does not include cash flows beyond payback period



Most useful for comparing projects with almost equal lives

Payback Period Example

<u>Project A</u>			<u>Project B</u>		<u>Project C</u>	
<u>Year</u>	<u>Cash Flow</u>	<u>Cumulative</u>	<u>Cash Flow</u>	<u>Cumulative</u>	<u>Cash Flow</u>	<u>Cumulative</u>
0	-\$1,000			-\$1,000	-\$1,000	
1	\$250	\$250	\$350	\$350	\$500	\$500
2	\$250	\$500	\$350	\$700	\$500	\$1,000
3	\$250	\$750	\$350	\$1,050	\$500	\$1,500
4	\$250	\$1,000	\$350	\$1,400		
5	\$250	\$1,250	\$350	\$1,750		
6	\$250	\$1,500				
7	\$250	\$1,750				
8	\$250	\$2,000				
9	\$250	\$2,250				
10	\$250	\$2,500				

Payback Period Comparison

<u>Project</u>	<u>Payback Period</u>	<u>Cash Return</u>
A	4 yrs.	\$2,500
B	3 (2.86) yrs.	\$1,750
C	2 yrs.	\$1,500

Discounted Payback Period

- Accounts for the time value of money
- Discount each future cash inflow to the present value
- Discount rate is the cost of capital or potential rate of return



<u>Year</u>	<u>Cash Flow</u>	<u>Discounted Cash Flow</u>	<u>Cumulative</u>	<u>Cash Flow</u>	<u>Discounted Cash Flow</u>	<u>Cumulative</u>	<u>Cash Flow</u>	<u>Discounted Cash Flow</u>	<u>Cumulative</u>
0	-\$1,000			-\$1,000			-\$1,000		
1	\$250	\$227	\$227	\$350	\$318	\$318	\$500	\$455	\$455
2	\$250	\$207	\$434	\$350	\$289	\$607	\$500	\$413	\$868
3	\$250	\$188	\$622	\$350	\$263	\$870	\$500	\$376	\$1,243
4	\$250	\$171	\$792	\$350	\$239	\$1,109			
5	\$250	\$155	\$948	\$350	\$217	\$1,327			
6	\$250	\$141	\$1,089						
7	\$250	\$128	\$1,217						
8	\$250	\$117	\$1,334						
9	\$250	\$106	\$1,440						
10	\$250	\$96	\$1,536						

Discounted Payback Period Example

Net Present Value (NPV)



PRESENT VALUE OF ALL CASH INFLOWS –
PRESENT VALUE OF ALL CASH
OUTFLOWS



ACCEPT ALL INVESTMENTS WITH A
POSITIVE NPV AND REJECT
INVESTMENTS WITH A NEGATIVE NPV

Net Present Value Example

		<u>Discounted</u> <u>Cash Flows</u>	<u>Discounted</u> <u>cash Flows</u>
<u>Year</u>	<u>Cash Flow</u>	<u>(i=5%)</u>	<u>(i=3%)</u>
0	-\$10,000	-\$10,000	-\$10,000
1	\$1,000	\$952	\$971
2	\$1,000	\$907	\$943
3	\$1,000	\$864	\$915
4	\$1,000	\$823	\$888
5	\$1,000	\$784	\$863
6	\$1,500	\$1,119	\$1,256
7	\$1,500	\$1,066	\$1,220
8	\$1,500	\$1,015	\$1,184
9	\$1,500	\$967	\$1,150
10	\$1,500	\$921	\$1,116
	NPV	-\$582	\$505

Profitability Index (PI)

- Divide the Present Value of cash inflows of the investment by the present value of cash outflows
- Accept project with $PI > 1$

Profitability Index – Example

<u>Year</u>	<u>Cash Flow</u>	<u>Discounted</u>	<u>Discounted</u>
		<u>Cash Flows</u>	<u>cash Flows</u>
		<u>(i=5%)</u>	<u>(i=3%)</u>
0	-\$10,000	-\$10,000	-\$10,000
1	\$1,000	\$952	\$971
2	\$1,000	\$907	\$943
3	\$1,000	\$864	\$915
4	\$1,000	\$823	\$888
5	\$1,000	\$784	\$863
6	\$1,500	\$1,119	\$1,256
7	\$1,500	\$1,066	\$1,220
8	\$1,500	\$1,015	\$1,184
9	\$1,500	\$967	\$1,150
10	\$1,500	\$921	\$1,116
	PV of cash		
	inflows	\$9,418	\$10,505
	PV of Cash		
	Outflow	\$10,000	\$10,000
	PI	94%	105%

Internal Rate of Return (IRR)

- Discount Rate that makes NPV equal to Zero
- Accept project if IRR is greater than threshold rate
- Reject project if IRR is lesser than threshold rate

Internal Rate of Return Example

<u>Year</u>	<u>Project A</u>	<u>Project B</u>
0	-\$10,000	-\$10,000
1	\$1,000	\$2,000
2	\$1,500	\$1,000
3	\$1,000	\$500
4	\$2,000	\$1,000
5	\$500	\$1,500
6	\$1,000	\$2,000
7	\$1,500	\$1,000
8	\$2,000	\$1,500
9	\$1,000	\$1,000
10	\$1,500	\$500
IRR	4.89%	3.69%